A SUBSET NOWHERE DENSE IN AN OPEN SUBSPACE IS NOWHERE DENSE IN THE WHOLE SPACE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The definition of a nowhere dense subset as one whose closure has empty interior, together with the behaviour of closures under passage to an open subspace. Set up in the author's section on nowhere dense sets, immediately before the counted statement. Fixes the property to be transferred and the closure computation used in the proof.

Lemma 0.1. *Let X be a topological space. Let $U \subset X$ be an open. Let $T \subset U$ be a subset. If T is nowhere dense in U, then T is nowhere dense in X.*

Proof. Assume T is nowhere dense in U. Suppose that $x \in X$ is an interior point of the closure $\overline{T}$ of T in X. Say $x \in V \subset \overline{T}$ with $V \subset X$ open in X. Note that $\overline{T} \cap U$ is the closure of T in U. Hence the interior of $\overline{T} \cap U$ being empty implies $V \cap U = \emptyset$. Thus x cannot be in the closure of U, a fortiori cannot be in the closure of T, a contradiction. □

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